

Table C1. Datamodel of the catalogues released with this work.

Column Name	Suffixes	Details	Units
K2_ID/ KIC_ID/ TESS_ID		K2/ <i>Kepler</i> / TESS identifier	
K2_Campaign		K2 campaign number (K2 Catalogues only)	
APOGEE_ID/ GALAH_ID		Spectroscopic survey identifier	
ASPCAP_ID		Identifies observation retained from APOGEE duplicates (APOGEE catalogues only, see Section 2.3.1)	
GaiaDR3_ID		<i>Gaia</i> DR3 identifier	
numax	add _err for uncertainty	Frequency of maximum oscillation power	μHz
Dnu	-'_'	Large frequency separation	μHz
EvoState		From Yu et al. (2018). RGB = 1, HeB = 2, unclassified = 0 (<i>Kepler</i> catalogue only)	
L	add _err for uncertainty	Luminosity based on <i>Gaia</i> parallax 17 μHz zeropoint correction used in K2 catalogues and Lindegren correction in <i>Kepler</i> and TESS (see Section 2.2)	$L_{\odot}$
Fe_H	add _err for uncertainty	[Fe/H] from APOGEE or GALAH	dex
alpha_M/ alpha_Fe	-'_'	$[\alpha/\text{M}]$ from APOGEE or $[\alpha/\text{Fe}]$ from GALAH	dex
Rg	add _16 for 16th or _84 for 84th percentile then add _BHM or _Gaia for source of distance (K2 BHM APOGEE catalogue only)	Guiding radius in MWPotential2014	kpc
Zmax	-'_'	Maximum vertical excursion in MWPotential2014	kpc
e	-'_'	Orbital eccentricity in MWPotential2014	
Lz	-'_'	Orbital angular momentum in MWPotential2014	kpc km s^{-1}
E	-'_'	Orbital energy in MWPotential2014	$\text{km}^2 \text{s}^{-2}$
vR	-'_'	Radial velocity component in MWPotential2014	km s^{-1}
vT	-'_'	Tangential velocity component in MWPotential2014	km s^{-1}
vZ	-'_'	Vertical velocity component in MWPotential2014	km s^{-1}
age	add _16 for 16th or _84 for 84th percentile then add _Dnu or _L for source of observational constraints (see Section 3.1.1)	Age from PARAM	Gyr
Mass	-'_'	Mass from PARAM	$M_{\odot}$
Radius	-'_'	Radius from PARAM	$R_{\odot}$
Dist	-'_'	Distance from PARAM	kpc
warn_low_numax		ν_{max} more than three standard deviations below 20 μHz	
warn_high_Dnu		$\Delta\nu \geq 21 \mu\text{Hz}$	
warn_high_RUWE		from <i>Gaia</i> DR3	
warn_NSS		-'_'	
warn_APOGEE/ warn_GALAH		Combination of flags from spectroscopy (see Section 2.3)	
drop_K2_dup		Duplicate target removed (K2 catalogues only)	
warn_plax_err		$\varpi/\sigma_{\varpi} < 5$	
warn_dustmaps		Validity of dustmap outputs (<i>Kepler</i> and K2 catalogues only)	
warn_nveAv	add _Dnu or _L for source of observational constraints	PARAM returns negative extinction	
warn_massDiff		Difference in mass from scaling relations based on L or $\Delta\nu$ exceeds 50% (See Section 3.1.1)	
Rel_data	add _Dnu or _L for source of observational constraints	combines flags from warn_low_numax to drop_K2_dup for $\Delta\nu$ datasets or from warn_low_numax to warn_dustmaps for L datasets to return stars with reliable data	
Rel_age	-'_'	combines Rel_data with warn_nveAv and warn_massDiff to return stars with reliable ages	